

Name:

Question 1 (1.2 points)

What is the difference between a pointer and a reference in C++? Give one example where a reference is preferable and one where a pointer is necessary.

Question 2 (1.2 points)

What is the purpose of a constructor and a destructor in a C++ class? Write a small class `Counter` that:

- Stores an integer.
- Initializes it in the constructor.
- Prints a message in the destructor.

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Question 3 (1.2 points)

What is inheritance in C++? Write a base class `Person` with a method `greet()`, and a derived class `Student` that overrides this method. The class `Student` should also store an attribute `grade`.

Question 4 (1.2 points)

What is a functor in C++? How does it differ from a lambda function? Provide an example.

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Question 5 (1.2 points)

What problem do smart pointers solve in C++? Briefly explain the difference between `std::unique_ptr` and `std::shared_ptr`.

Question 6 (1.2 points)

In Python, what is the difference between a list and a tuple? Give one example where a tuple is more appropriate than a list.

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Question 7 (1.2 points)

Write a Python function `square_even(numbers)` that

- Takes a list of integers.
- Returns a new list containing the squares of only the even numbers.

You may use a loop or a list comprehension.

Question 8 (1.2 points)

Write a Python class `BankAccount` with:

- An attribute `balance`.
- A method `deposit(amount)`.
- A method `withdraw(amount)` that prevents overdrawing.

Then explain how you would create multiple account objects.

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Question 9 (1.2 points)

Using NumPy:

- Create a 1D array with values from 0 to 9.
- Select only the even elements.
- Compute their mean.

Explain why this approach is efficient compared to a pure Python loop.

Question 10 (1.2 points)

You want to expose the following C++ class to Python using pybind11:

```
class Calculator {  
public:  
    int add(int a, int b);  
};
```

Write a minimal PYBIND11_MODULE binding code. Explain how an object of this class would be created and used in Python.